

Pre-Move Detection System

Pipeline & Strategy Guide

NSE Equities - Order Flow & Microstructure Analysis

Model: LightGBM Multiclass | Label: 60m 1% Triple-Barrier

Data: Tick + DOM Level 2 | Universe: ~483 Symbols | Period: Jun 3-12, 2026

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1. Feature Importance & Weights

The LightGBM multiclass model (3 classes: LONG, SHORT, NO_TRADE) was trained on 18 microstructure and trade-flow features. Importance is measured by total gain (improvement in loss when a feature is used for splitting), aggregated across all trees. Higher gain = more predictive power.

Top 18 Features by Gain Importance

#	Feature	Gain	%	Splits	%	Range (min..	
1	delta_1m	103,263	19.9%	1,660	18.4%	-32787484.0..	26604
2	iceberg_score	88,268	17.0%	371	4.1%	0.0..100.0	
3	volatility_5m	66,730	12.9%	833	9.3%	0.0..333.2	
4	large_trade_ratio	53,986	10.4%	963	10.7%	0.0..1.0	
5	vwap_distance	47,923	9.2%	642	7.1%	-0.0..0.0	
6	aggressor_ratio	44,664	8.6%	1,102	12.2%	0.0..1.0	
7	spread	37,488	7.2%	1,004	11.2%	0.0..337.8	
8	trade_count_burst	30,245	5.8%	591	6.6%	0.0..14.4	
9	depth_drop_ask	8,506	1.6%	270	3.0%	-26467342.0..	25267
10	volume_burst	7,508	1.4%	290	3.2%	0.0..19.9	
11	imbalance_top5	6,595	1.3%	338	3.8%	-1.0..1.0	
12	bid_replenishment_	5,722	1.1%	129	1.4%	0.0..1.0	
13	absorption_buyer_1	4,436	0.9%	168	1.9%	0.0..0.0	
14	absorption_seller_	3,015	0.6%	125	1.4%	0.0..0.0	
15	absorption_buyer_5	2,873	0.6%	122	1.4%	0.0..0.0	
16	price_acceleration	2,676	0.5%	158	1.8%	-490.2..707.8	
17	depth_drop_bid	2,346	0.5%	119	1.3%	-21717456.0..	14823
18	absorption_seller_	1,938	0.4%	115	1.3%	0.0..0.0	

Top 8 features: 91.2% cumulative gain importance

Bottom 10 features: 8.8% cumulative gain importance

Key Observations

- delta_1m (net order flow) is the single most powerful feature at 19.9% gain
- iceberg_score is ACTIVE (min=0, max=100, avg=81.4, 312K unique values) -- contrary to earlier notes
- Absorption features (buyer_1m/5m, seller_1m/5m) are near-zero but the model still extracts marginal signal
- depth_drop_bid is the weakest feature at 0.5% gain -- limited predictive value
- The 4 absorption + 2 depth_drop + price_acceleration + bid_replenish = 8.8% combined -- removable without material impact

2. Per-Tick Processing

Every incoming trade tick updates a set of running counters and state variables. No model inference happens at tick level -- that would be too expensive (483 symbols x hundreds of ticks/min). The tick-level processing is purely arithmetic aggregation.

What Updates on Every Tick

Variable	Update Rule	Purpose
ltp	Overwrite with trade price	Latest price for features
VWAP	Running ($\text{sum}(\text{price} \times \text{vol}) / \text{sum}(\text{vol})$)	VWAP distance feature
volume_1m	$\text{volume_1m} += \text{trade_qty}$	Volume burst calculation
buy_volume_1m	$+= \text{qty}$ if aggressive buy	$\text{delta_1m} = \text{buy} - \text{sell}$
sell_volume_1m	$+= \text{qty}$ if aggressive sell	$\text{delta_1m} = \text{buy} - \text{sell}$
trade_count_1m	$+= 1$	Trade count burst
large_trade_count	$+= 1$ if $\text{qty} > 10 \times \text{median}$	Large trade ratio
max_price, min_price	Update running high/low	Volatility feature

DOM snapshots (when separate stream arrives) likewise update: best bid/ask prices, total bid/ask quantities at levels 1-5, spread, depth drops. These are also aggregated to 1-minute averages -- not used at tick-level for inference.

Pseudo-Code: Tick Handler

```
def on_tick(tick):
    ltp = tick.price
    vwap_numer += tick.price * tick.qty
    vwap_denom += tick.qty
    volume_1m += tick.qty
    trade_count_1m += 1
    if tick.is_aggressive_buy:
        buy_volume_1m += tick.qty
    elif tick.is_aggressive_sell:
        sell_volume_1m += tick.qty
    if tick.qty > median_qty * 10:
        large_trade_count += 1
    max_price_1m = max(max_price_1m, tick.price)
    min_price_1m = min(min_price_1m, tick.price)
    # No model inference here
```

These counters are reset at the start of each 1-minute window. Note: This describes the online (live) path. For historical processing, the same aggregation is done in batch via DuckDB queries.

3. Minute-End Aggregation & Feature Computation

At the end of each 1-minute bar, all tick-level counters and DOM snapshots within that minute are aggregated into a single feature vector. This is where model inference happens.

DOM Resampling (DuckDB)

DOM snapshots within the minute are averaged via DuckDB SQL. For a unified feed (DOM columns embedded in tick data), the tick dataframe serves as the DOM source directly:

```
SELECT date_trunc('minute', ts) AS ts,
       AVG(ltp) AS ltp,
       LAST(ltp) AS close_price,
       AVG(bqty1), AVG(bqty2), ..., AVG(bqty5),
       AVG(aqty1), AVG(aqty2), ..., AVG(aqty5),
       AVG(bprice1), AVG(aprice1),
       AVG(bqty1 - aqty1) AS spread
FROM dom_source
GROUP BY date_trunc('minute', ts)
```

Feature Computation per Minute

Once the 1-minute bar is formed, the 18 features are computed from the aggregated data:

Feature	Formula / Source	Group
delta_1m	buy_volume_1m - sell_volume_1m	Trade Flow
volume_burst	volume_1m / rolling_20m_avg_volume	Trade Flow
aggressor_ratio	aggressive_buys / total_trades	Trade Flow
trade_count_burst	trade_count_1m / rolling_20m_avg_count	Trade Flow
large_trade_ratio	large_trades / total_trades	Trade Flow
imbalance_top5	(bidqty1-5 - askqty1-5) / (bidqty1-5 + askqty1-5)	Depth
spread	ask1_price - bid1_price	Depth
depth_drop_bid	change in total bid qty vs prev minute	Depth
depth_drop_ask	change in total ask qty vs prev minute	Depth
vwap_distance	(ltp - vwap) / vwap	Price
volatility_5m	stddev(close_price) over 5 min	Price
price_acceleration	2nd derivative of close_price	Price
iceberg_score	executed_qty / displayed_qty	Micro
bid_replenishment_	bid qty recovered / bid qty consumed	Micro
absorption_*	passive fill qty / aggressive qty (1m/5m)	Micro

Model Inference

With the 18-dimensional feature vector computed, the LightGBM model produces 3 probabilities: prob_long, prob_short, prob_no_trade. The 9-period ROC of close_price is also pre-calculated continuously across day boundaries (no NaN at market open).

```
minute_end_vector = [delta_1m, volume_burst, aggressor_ratio, ..., absorption_seller_5m]
prob_long, prob_short, prob_no_trade = model.predict(minute_end_vector)
roc_9 = (close_price - close_price[-9]) / close_price[-9] * 100
```

4. Threshold Confirmation & Trigger Logic

The model probabilities from each 1-minute bar are evaluated against configurable thresholds to generate trading alerts. The current active configuration uses asymmetric thresholds.

Alert Generation Rules

LONG Alert	prob_long >= 0.30 AND prob_long > prob_short
SHORT Alert	prob_short >= 0.25 AND prob_short > prob_long
NO_TRADE	Neither condition met
Dominance Check	The predicted direction must have HIGHER probability than the opposite
Alert Expiry	Alert expires after 360 minutes without confirmation
Direction Cancel	Opposite alert cancels the current alert (last-wins)

Pseudo-Code: Alert Decision

```
def evaluate_alert(p_l, p_s, theta_long=0.30, theta_short=0.25):
    if p_l >= theta_long and p_l > p_s:
        return 'LONG'
    elif p_s >= theta_short and p_s > p_l:
        return 'SHORT'
    else:
        return 'NO_TRADE'

# Alert state machine:
if alert_direction == 'LONG' and alert_age > 360:
    alert_active = False # expired
if new_alert != alert_direction:
    alert_direction = new_alert # flip direction
```

The probability thresholds (0.30 LONG, 0.25 SHORT) were selected through a systematic sweep in 75_tune_lightgbm_threshold.py. 0.30 was chosen for LONG as the best trade-off between precision (17.9%) and recall (26.4%). At lower thresholds like 0.20, precision drops to 9.7% with 3.3x more false signals.

5. ROC Confirmation Gate

After a threshold alert fires, the system waits for a Rate-of-Change (ROC) confirmation before entering a trade. This prevents entering on stale alerts where price has reversed.

ROC Definition

$$\text{ROC}_9(t) = (\text{close_price}[t] - \text{close_price}[t-9]) / \text{close_price}[t-9] * 100$$

The 9-period ROC uses close_price (1-minute bars), so it covers approximately 9 minutes of price action. It is pre-calculated continuously across day boundaries -- meaning the last 9 bars of the previous trading day carry forward. This eliminates the early-morning NaN blindspot.

Confirmation Rules

LONG Confirmation	ROC_9 >= +1.0%
SHORT Confirmation	ROC_9 <= -1.0%
Must follow alert	Alert must have fired first (prob threshold)
Alert-to-ROC distance	Tracked but no max gap -- any alert within 360 min qualifies
Edge case	If alert fires and ROC already satisfied at same minute -> immediate entry

Why ROC and not raw probability?

The model probability is a static assessment of the current 1-minute bar. ROC adds momentum confirmation -- are prices actually moving in the predicted direction? This eliminates cases where the model flags a setup but the stock fails to follow through. In practice, ROC filters out approximately 30-40% of raw threshold alerts, improving win rate significantly.

6. Daily Open Filter & Bypass Exceptions

Daily Open Trend Filter

Before executing the trade, the system checks whether the current price is on the correct side of the day's opening price. This is a trend-continuity filter.

LONG	entry_price > daily_open (price above open)
SHORT	entry_price < daily_open (price below open)
Rationale	Stocks that are above open tend to stay above; below open tend to stay below
Limitation	Stocks opening near flat can oscillate around the open, causing early exits

Bypass Exceptions (Asymmetric)

When the Daily Open filter fails, bypass exceptions can still allow the trade. These are asymmetric -- enabled for LONG, disabled for SHORT -- based on empirical testing.

Direction	Bypass	Rule 1 (High Prob)	Rule 2 (Extreme Z)
LONG	ENABLED	prob_long >= 0.35	imbalance_top5_z < -1.5
SHORT	DISABLED	N/A	N/A (strict filter always)
Z-score			

Cross-sectional Z of imbalance_top5 and aggressor_ratio across all ~483 symbols at current minute

Why SHORT Bypass is Disabled

The SHORT bypass was tested and rejected because it generated too many fake trades. With bypass enabled at 0.25 SHORT: +11 additional movers captured but at a cost of +122 fake trades (11:1 ratio). The LONG bypass, in contrast, adds +9 movers at +36 fakes (4:1 ratio), which is considered tradeable. Disabling SHORT bypass also automatically blocks dangerous trades like the AEGISLOG SHORT on June 11 which suffered -15% MAE.

Z-Score Gate Rules (after bypass, before entry)

After passing ROC and Daily Open (or bypass), the trade must satisfy one of two Z-score paths. These are cross-sectional Z-scores computed across all symbols at each minute.

Path	Condition	LONG	SHORT
Steady Flow	Turnover	delta_value_lakhs > 30	delta_value_lakhs < -30
	Block Filter	large_trade_ratio <= 0.04	large_trade_ratio <= 0.04
	Z-Imbalance	imbalance_top5_z < 0.10	imbalance_top5_z > -0.10
	Z-Aggressor	aggressor_ratio_z >= 0.0	aggressor_ratio_z <= 0.0
Block Pullback	Turnover	delta_value_lakhs > 50	delta_value_lakhs < -50
	Block Filter	0.04 < ltr <= 0.10	0.04 < ltr <= 0.10
	VWAP Dist	vwap_distance <= 0 (below)	vwap_distance >= 0 (above)
	Z-Imbalance	imbalance_top5_z < 0.10	imbalance_top5_z > -0.10
	Z-Aggressor	aggressor_ratio_z >= 0.0	aggressor_ratio_z <= 0.0

delta_value_lakhs = delta_1m x entry_price / 100,000. This standardizes share-count turnover into a cross-stock-comparable rupee-value measure. The two paths (Steady Flow and Block Pullback) cover different market microstructures -- gradual accumulation vs. institutional block absorption.

7. End-to-End Trade Decision Flow

The complete flow from tick arrival to trade entry. Each step is a hard gate -- if the condition fails, the trade is blocked (unless bypass applies).

1	Tick Arrives	Update running counters (ltp, VWAP, volume, delta, etc.)
2	Minute Boundary	Aggregate tick + DOM -> 18 features. Model inference -> prob_long, prob_short
3	Threshold Gate [GATE]	prob >= 0.30 (LONG) or >= 0.25 (SHORT) AND dominant over opposite? If NO -> BLOCK
4	ROC Confirmation [GATE]	ROC_9 >= +1% (LONG) or <= -1% (SHORT)? If NO -> BLOCK (alert stays active)
5	Daily Open Filter [GATE]	entry_price > daily_open (LONG) or < daily_open (SHORT)?
6	Bypass Check [GATE]	If Daily Open FAILED: LONG checks bypass. SHORT always blocked.
7	Z-Score Gates [GATE]	Must satisfy Steady Flow OR Block Pullback path
8	Deduplication [GATE]	If same symbol-day-direction already traded -> BLOCK (separate per direction)
9	TRADE ENTRY	Enter at current LTP. Exit at EOD (no intraday exits)

A. Appendix: Quick Reference

Active Configuration Summary

Model	LightGBM Multiclass (LONG/SHORT/NO_TRADE)
Label	label_60m_1pct (triple-barrier, 60min, 1%)
Features	18 microstructure + trade-flow features
LONG Threshold	0.30
SHORT Threshold	0.25
ROC Confirmation	9-period, >=+1% LONG, <=-1% SHORT
Daily Open Filter	LONG: price > open, SHORT: price < open
LONG Bypass	prob >= 0.35 OR imbalance_top5_z < -1.5
SHORT Bypass	DISABLED (strict filter always)
Z-Score Gates	Steady Flow OR Block Pullback
Deduplication	Separate per direction
Exit	EOD only, no intraday/overnight
Total Trades (8 days)	323 (184 LONG + 139 SHORT)
Avg Trades/Day	~40
Win Rate	~28% (EOD MFE >= 3%)

Feature Groups at a Glance

Trade Flow (5)	delta_1m, volume_burst, aggressor_ratio, trade_count_burst, large_trade_ratio
Depth (4)	imbalance_top5, spread, depth_drop_bid, depth_drop_ask
Price (3)	vwap_distance, volatility_5m, price_acceleration
Microstructure (6)	iceberg_score, bid_replenish_rate, absorption_buyer_1m/5m, seller_1m/5m

Key Files

Model Weights	models/lgbm_model_60m_1pct_final.txt
OOF Predictions	results/oof_predictions_lgbm_label_60m_1pct.parquet
Final Trades	results/performance_sweeps.xlsx -> Recommended_Trades_Log
All Sweeps	results/all_triggered_trades.csv (9,279 rows)
Strategy Spec	docs/HANDOVER_STRATEGY_REPORT.md
Feature Engine	features/feature_factory.py
Trigger Sim	scripts/75_tune_lightgbm_threshold.py